

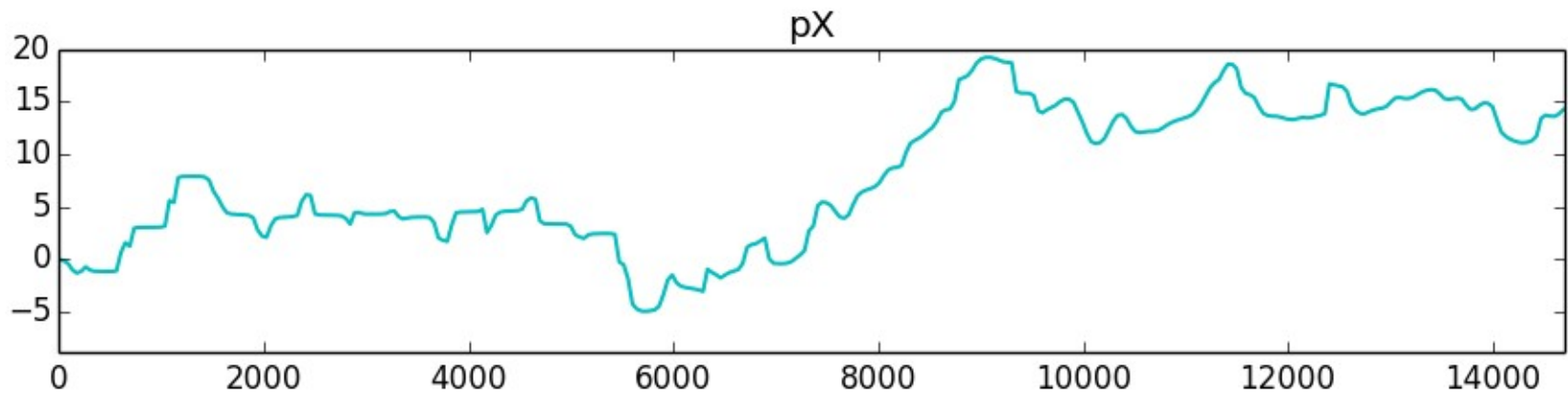
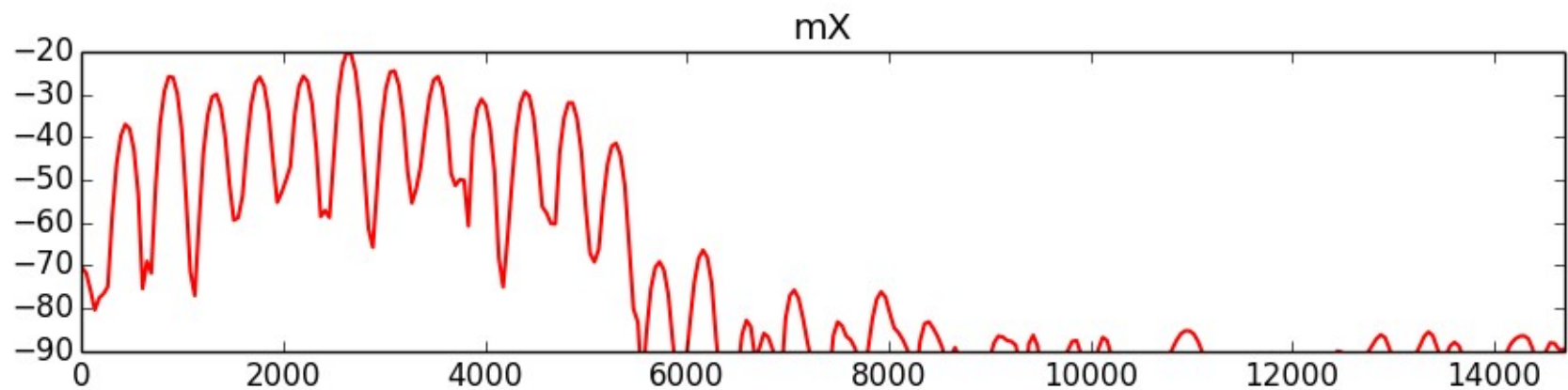
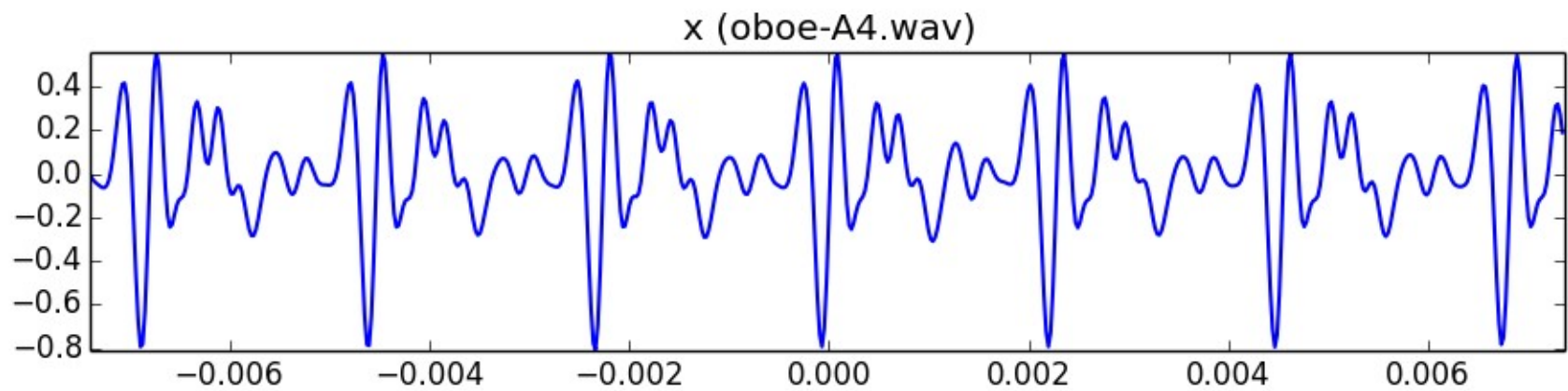
6T2: Fundamental frequency detection

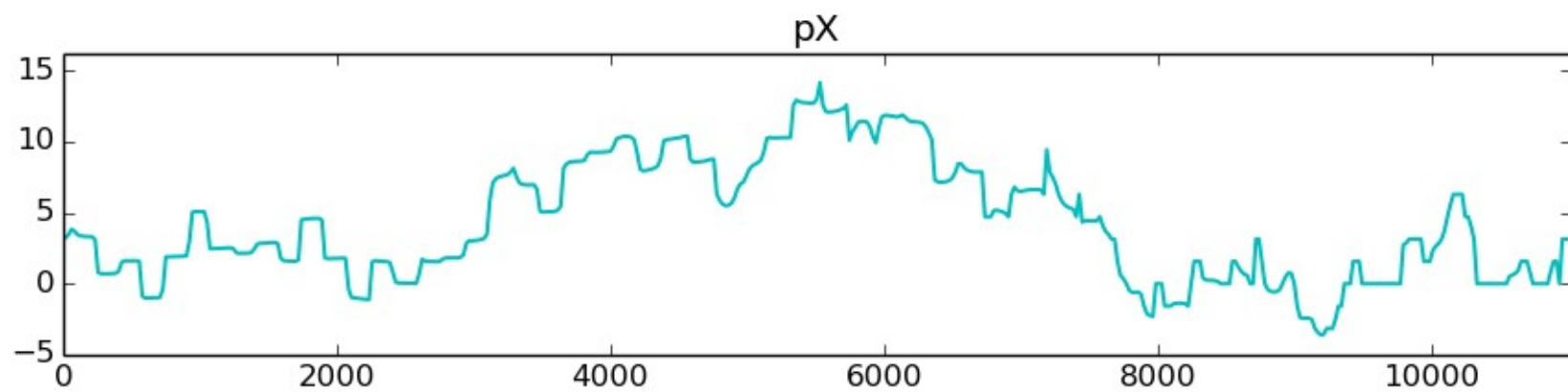
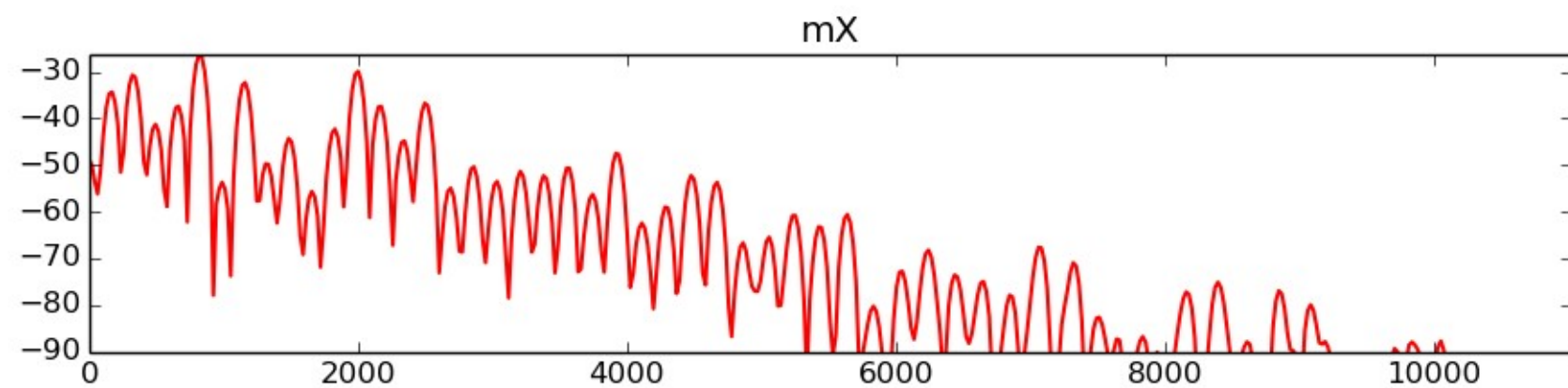
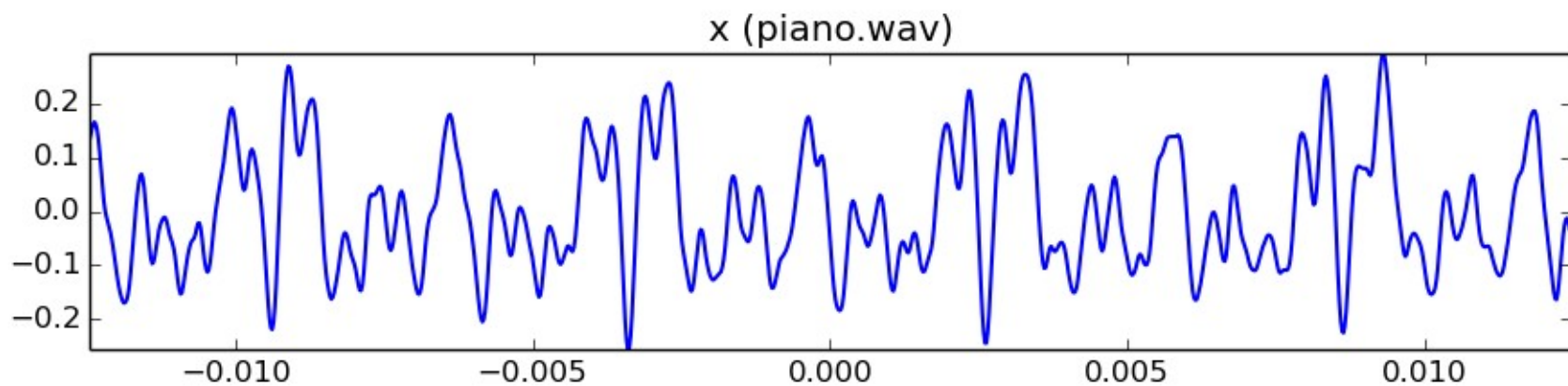
Xavier Serra

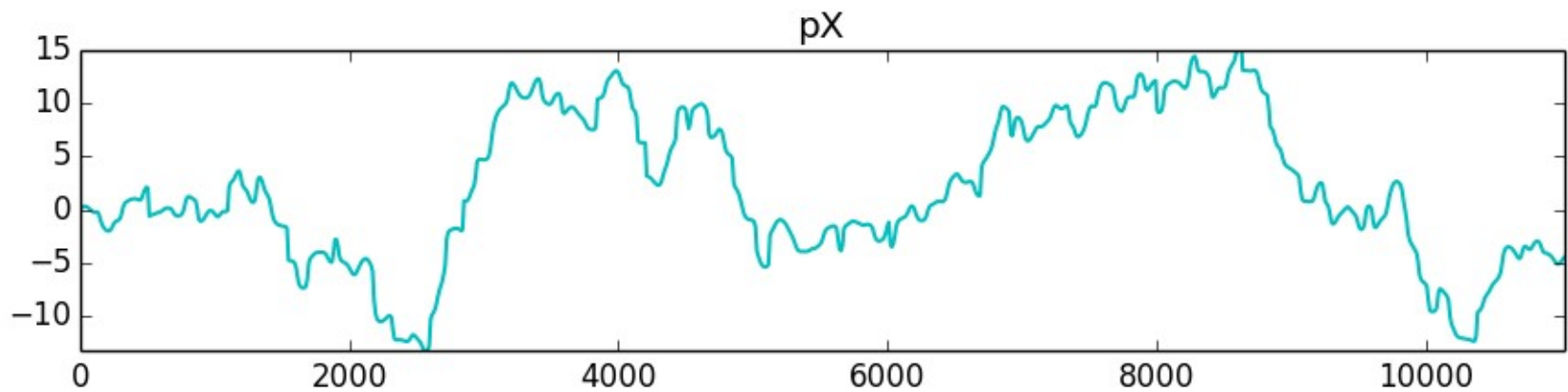
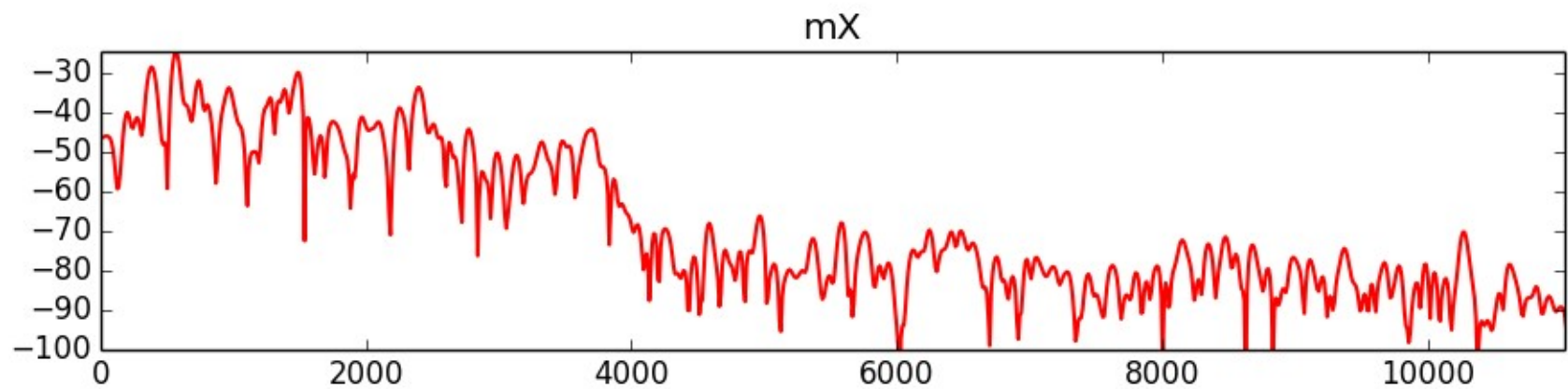
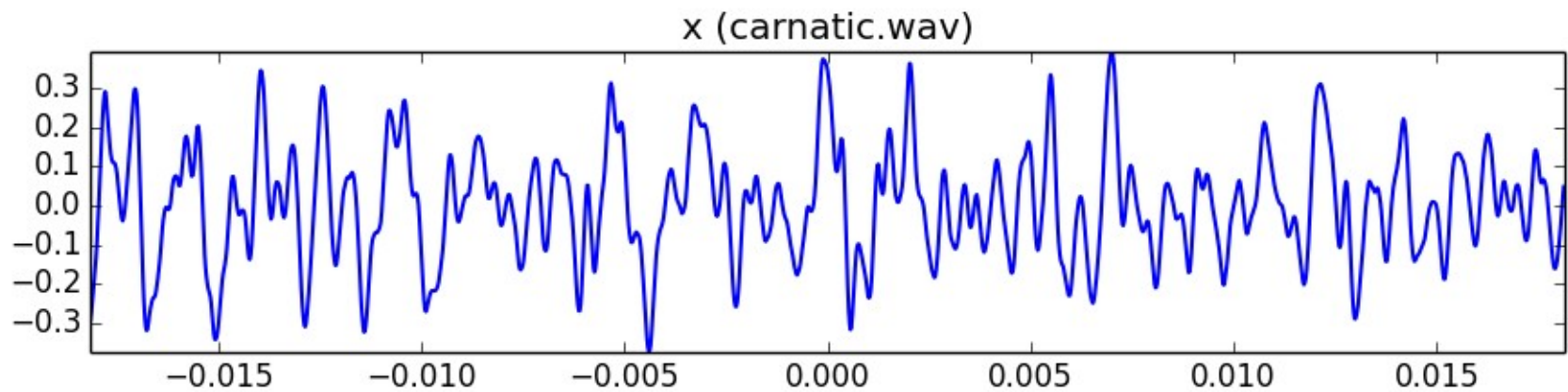
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- F0 detection in time domain
 - monophonic signals
- F0 detection in frequency domain
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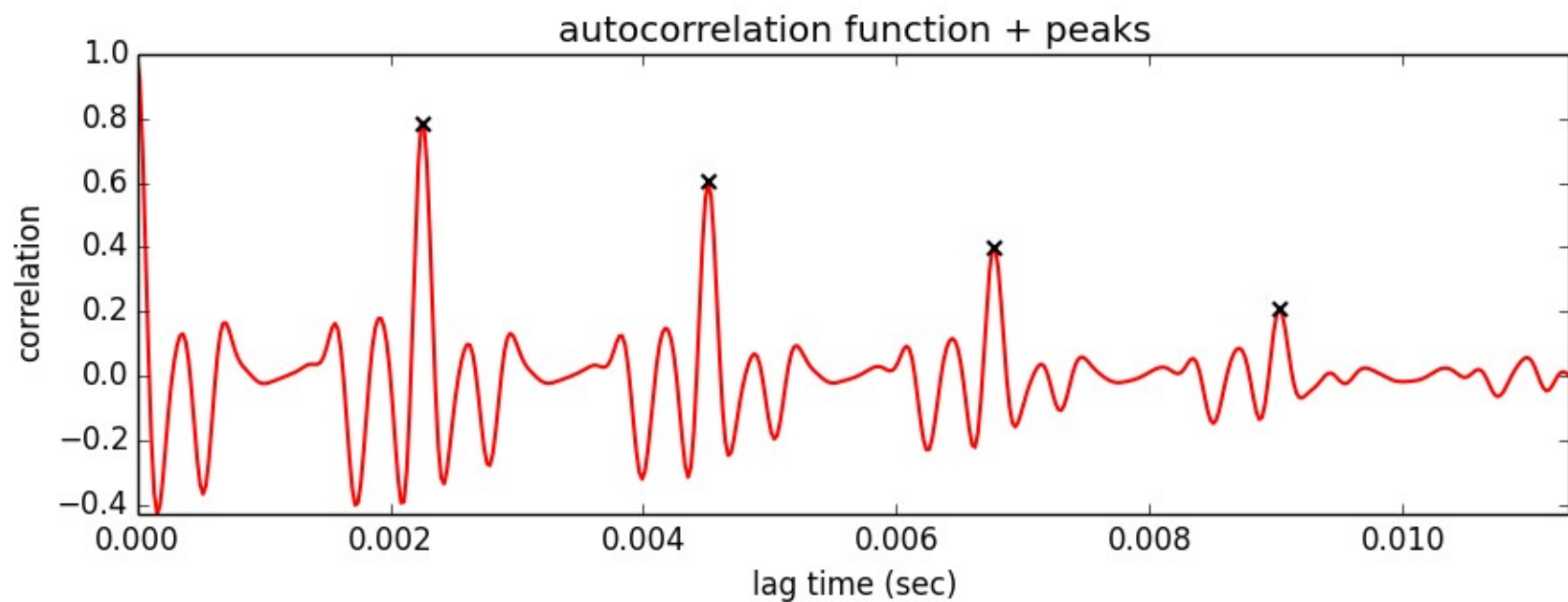
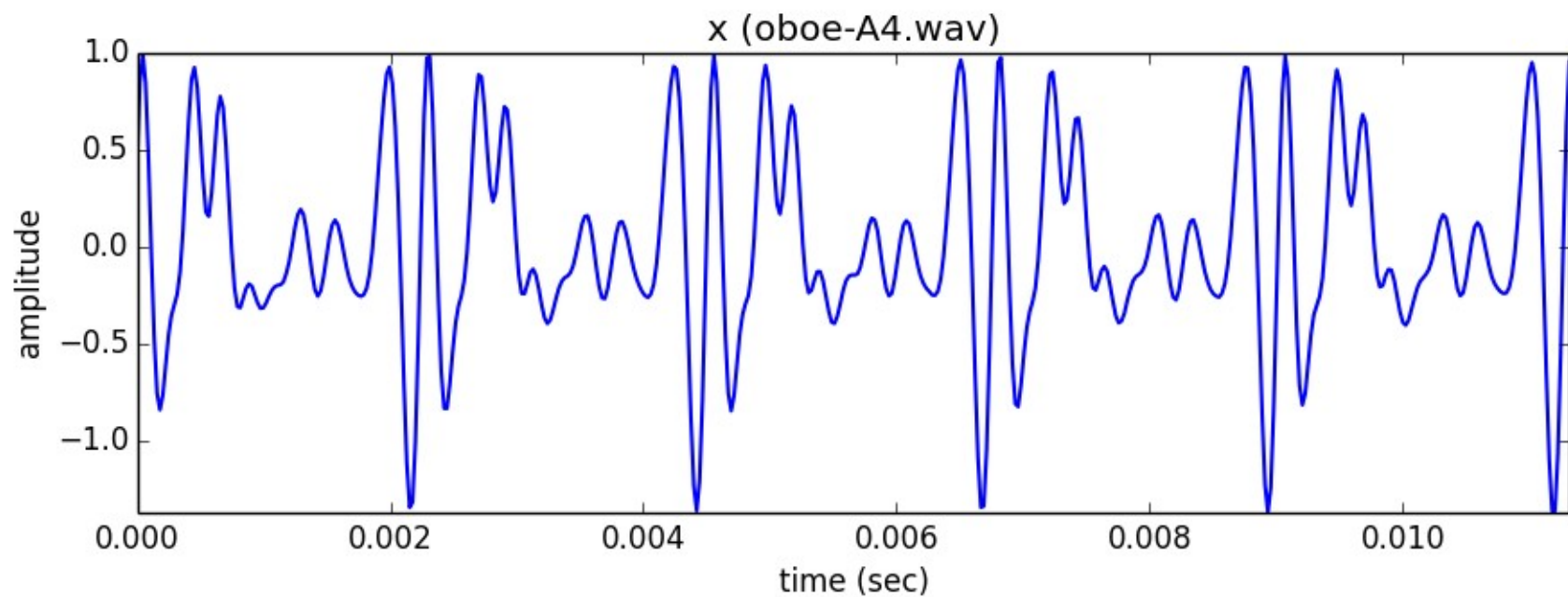


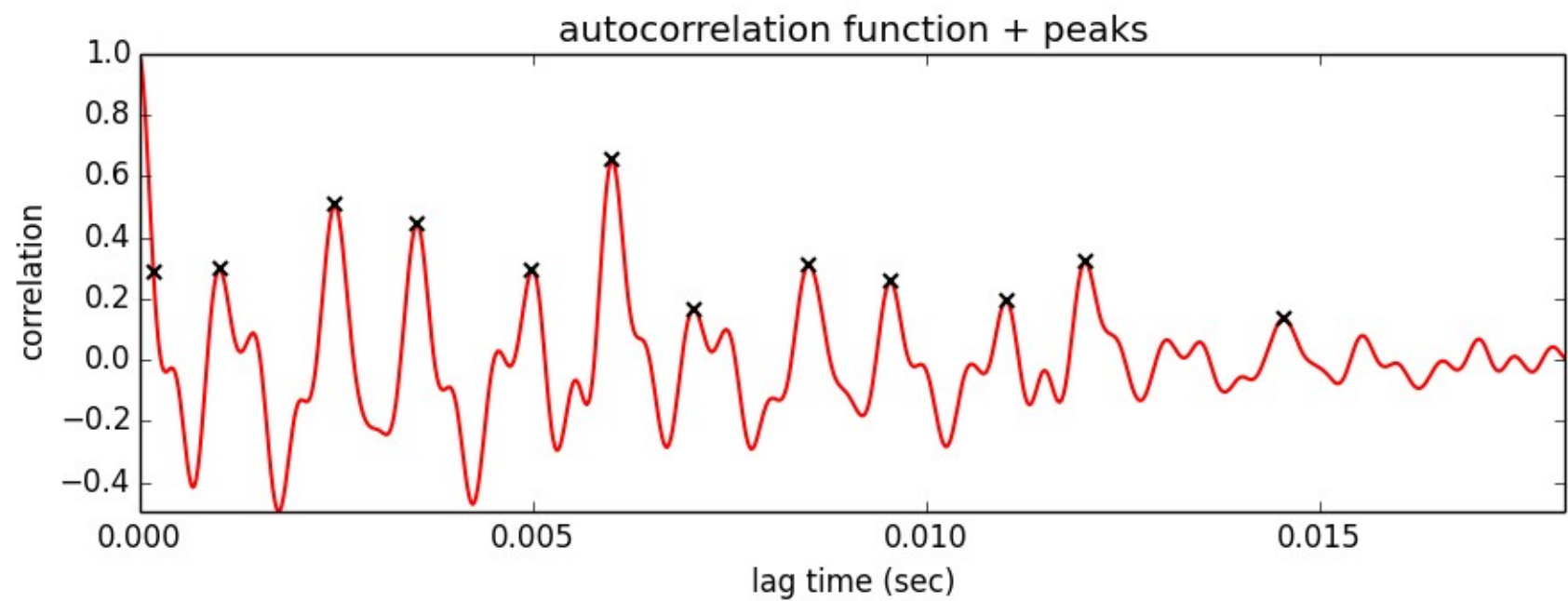
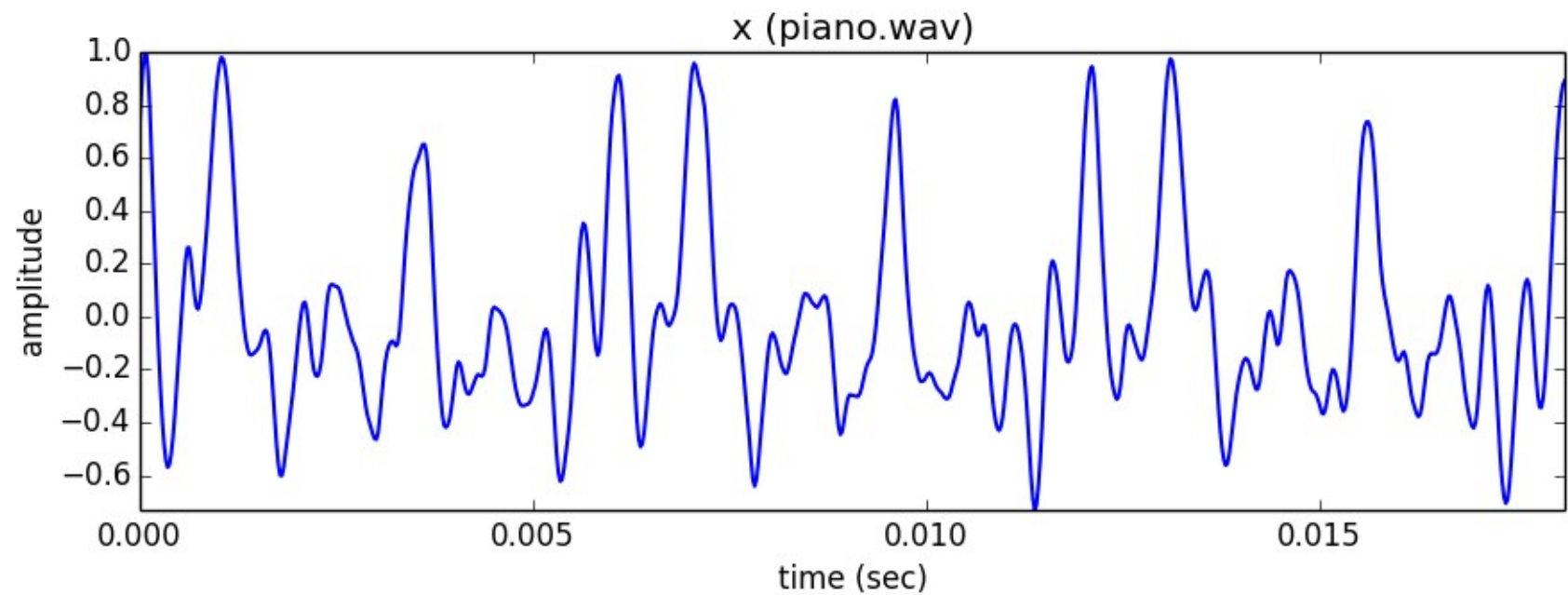
F0 detection in time domain

Autocorrelation function (with tapering)

$$r_x[l] = \sum_{n=0}^{N-1-l} x[n]x[n+l] \quad l=0,1,\dots,N-1$$

where $l = \text{lag}$



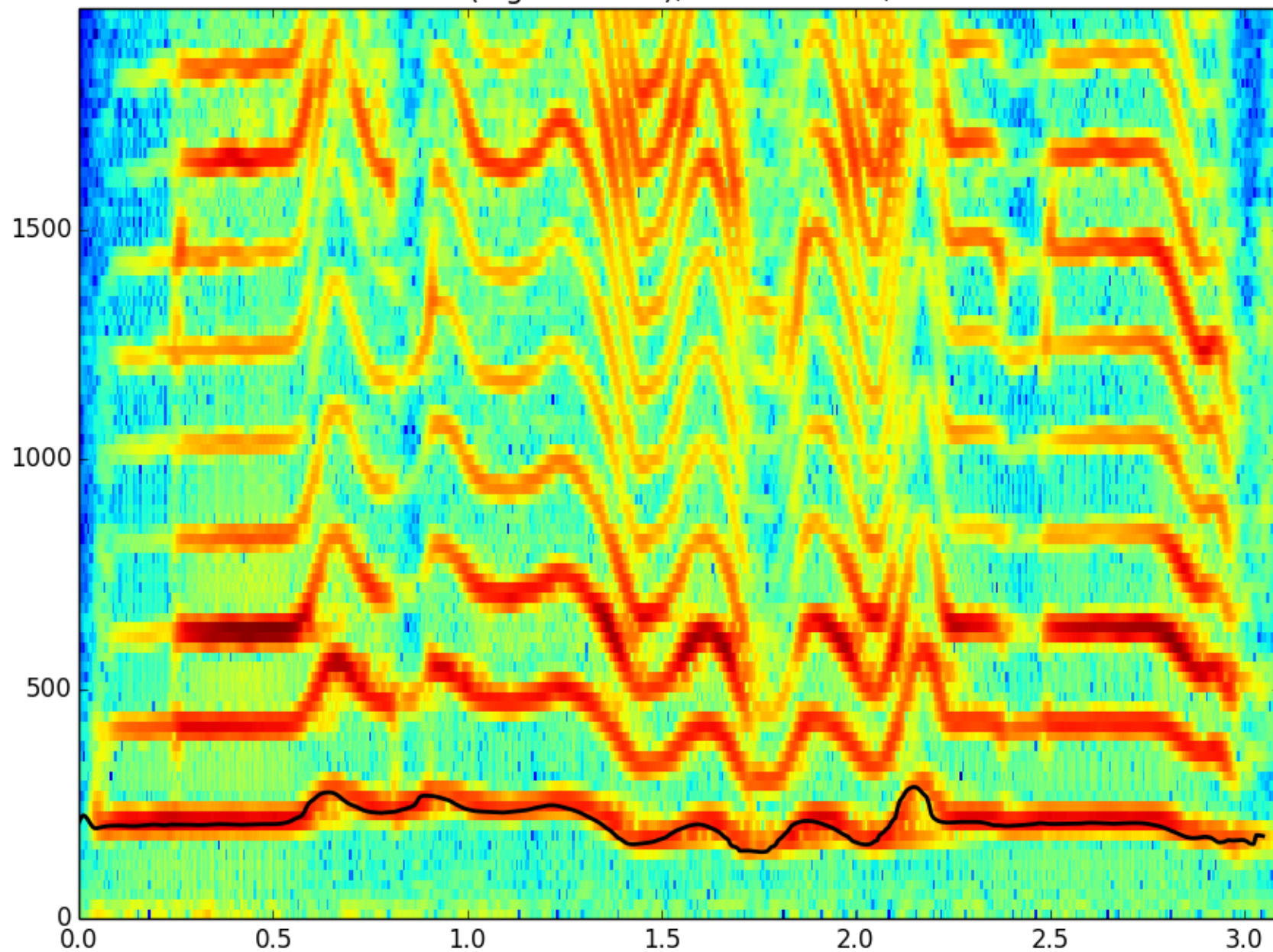


YIN Algorithm (Cheveigné and Kawahara, 2002)

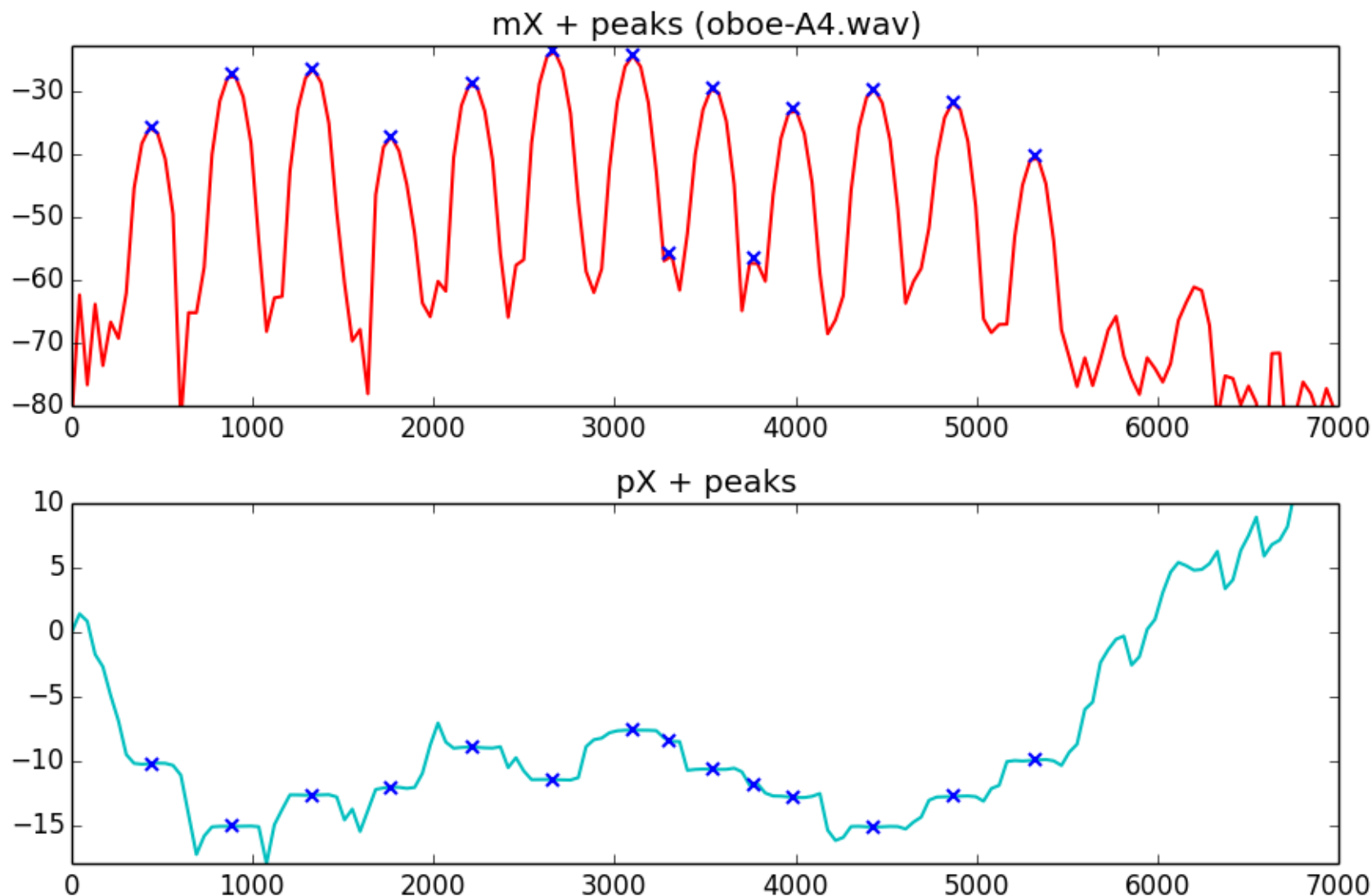
- Based on the difference function

$$d[l] = \sum_{n=0}^{n=N-1-l} (x[n] - x[n+l])^2 \quad l=0,1,\dots,N-1$$

mX + f0 (vignesh.wav), YIN: N=2048, H = 256

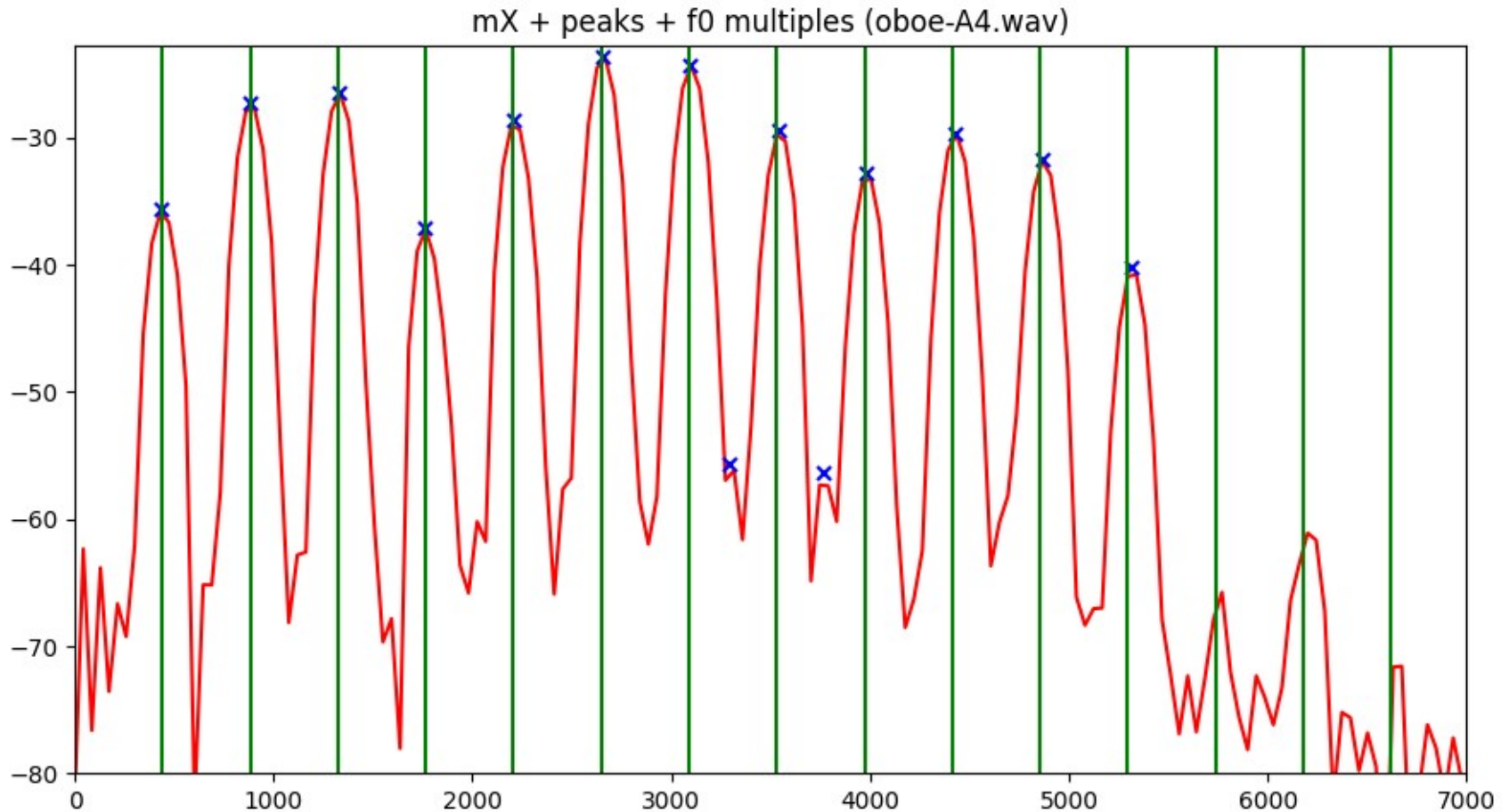


F0 detection in frequency domain



F0 in the spectrum

The F0 can be defined as the common divisor of the harmonic series that best explains the spectral peaks.



Pattern matching

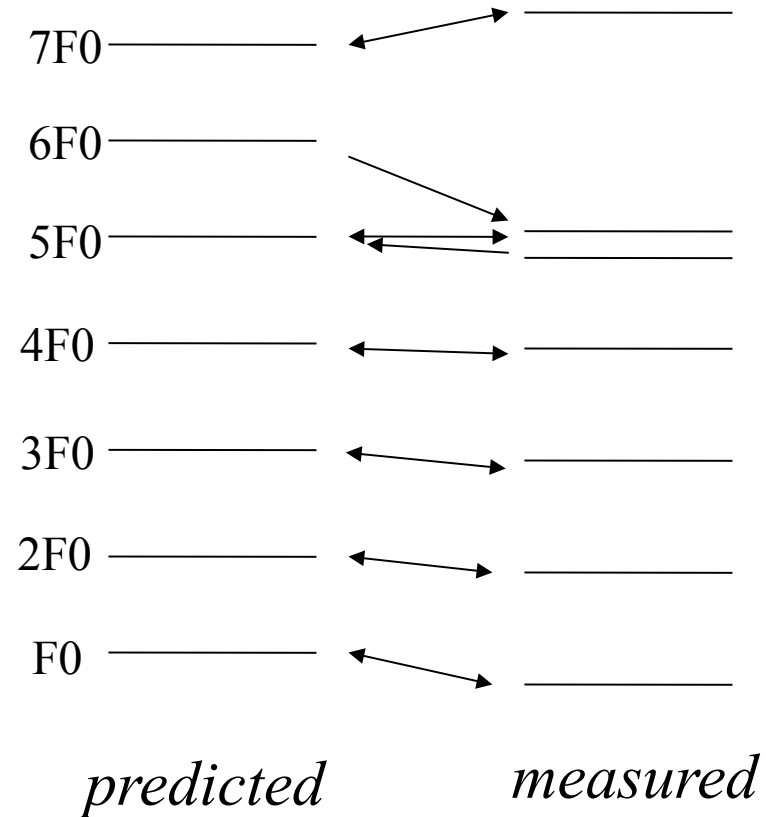
Two-way mismatch algorithm (Maher and Beauchamp, 1994)

$$\begin{aligned}\text{Err}_{p \rightarrow m} &= \sum_{n=1}^N E_{\omega}(\Delta f_n, f_n, a_n, A_{\max}) \\ &= \sum_{n=1}^N \Delta f_n \cdot (f_n)^{-p} \\ &\quad + \left(\frac{a_n}{A_{\max}} \right) \times [q \Delta f_n \cdot (f_n)^{-p} - r]\end{aligned}$$

Δf_n : diff. between predicted and
the closest measured peaks

f_n, a_n : frequency and magnitude of
predicted peaks

$A_{\max}$: maximum peak magnitude



$$\begin{aligned}\text{Err}_{m \rightarrow p} &= \sum_{k=1}^K E_{\omega}(\Delta f_k, f_k, a_k, A_{\max}) \\ &= \sum_{k=1}^K \Delta f_k \cdot (f_k)^{-p} + \left(\frac{a_k}{A_{\max}}\right) \times [q \Delta f_k \cdot (f_k)^{-p} - r]\end{aligned}$$

Δf_k : diff. between predicted and its closest measured peaks

f_k, a_k : frequency and magnitude of predicted peaks

$A_{\max}$: maximum peak magnitude

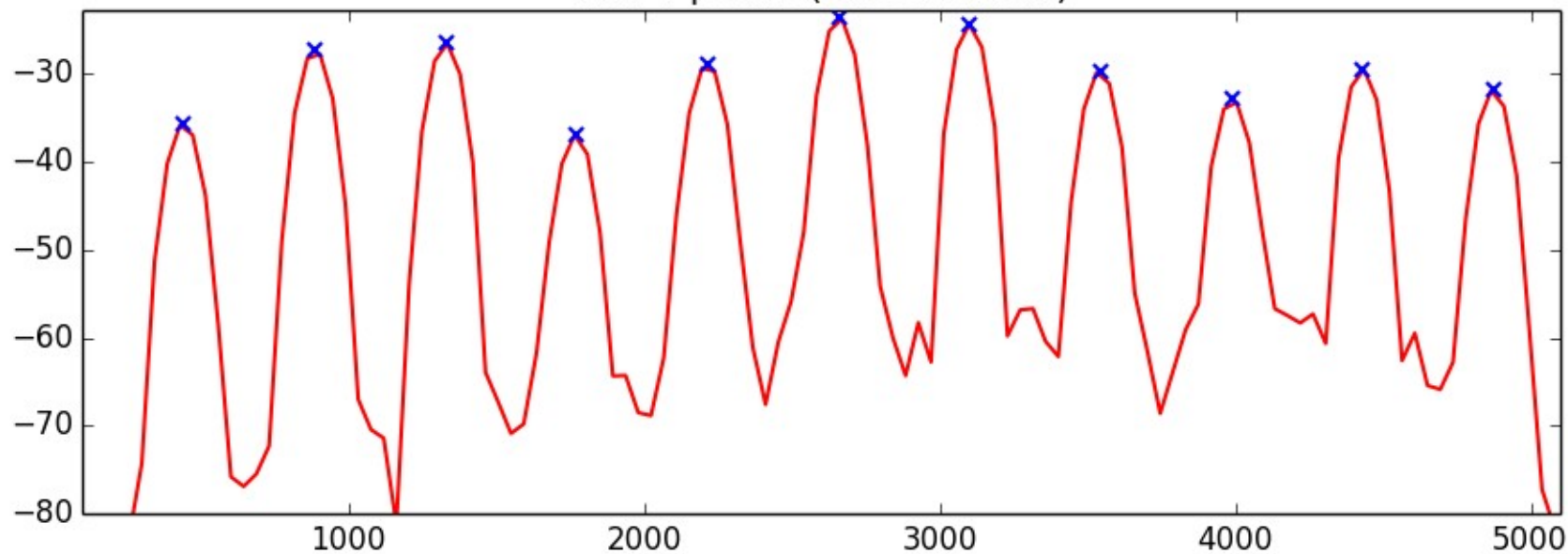
Total error: $\text{Err}_{\text{total}} = \text{Err}_{p \rightarrow m} / N + \rho \text{Err}_{m \rightarrow p} / K$

Maher and Beauchamp propose: $p = 0.5, q = 1.4, r = 0.5, \rho = 0.33$

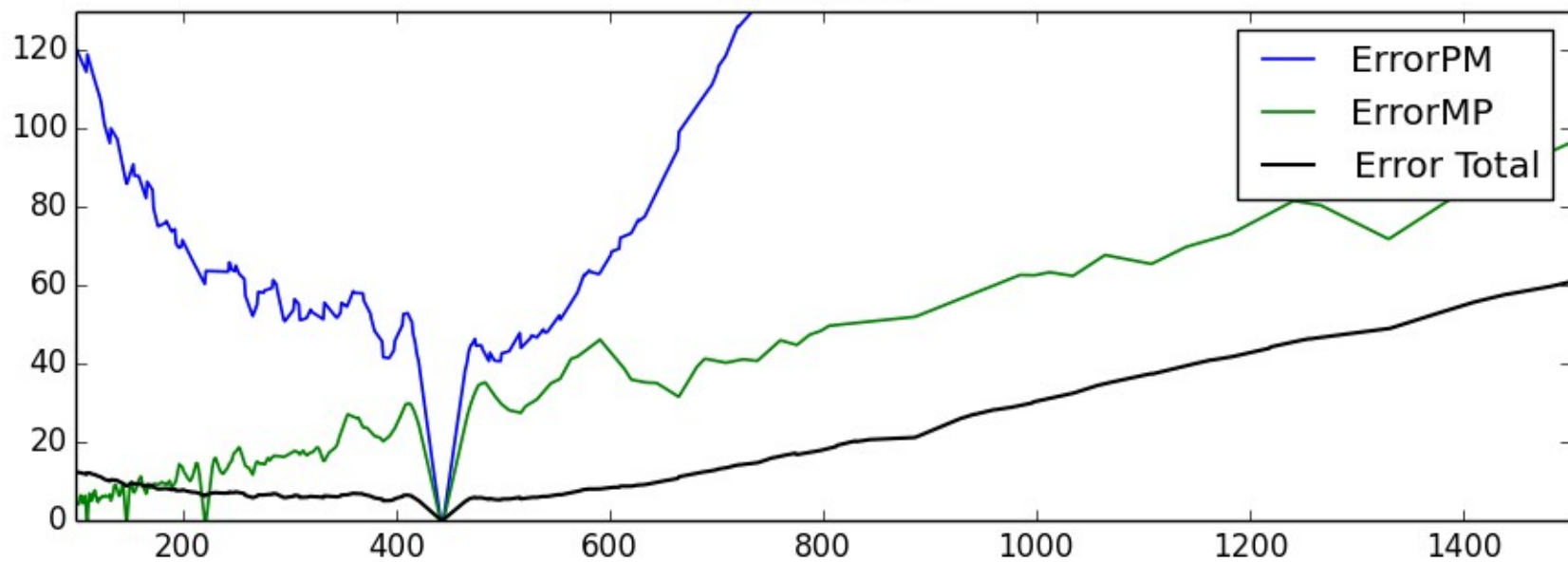
	$\text{Err}_{p \rightarrow m}$	$\text{Err}_{m \rightarrow p}$	$\text{Err}_{\text{total}}$
50Hz	122.58	-3.0	7.49
100Hz	32.0	-3.0	3.83
200Hz	10.0	30.66	4.2

TWM error calculation from the frequencies: 200, 300, 500, 600, 700, 800.

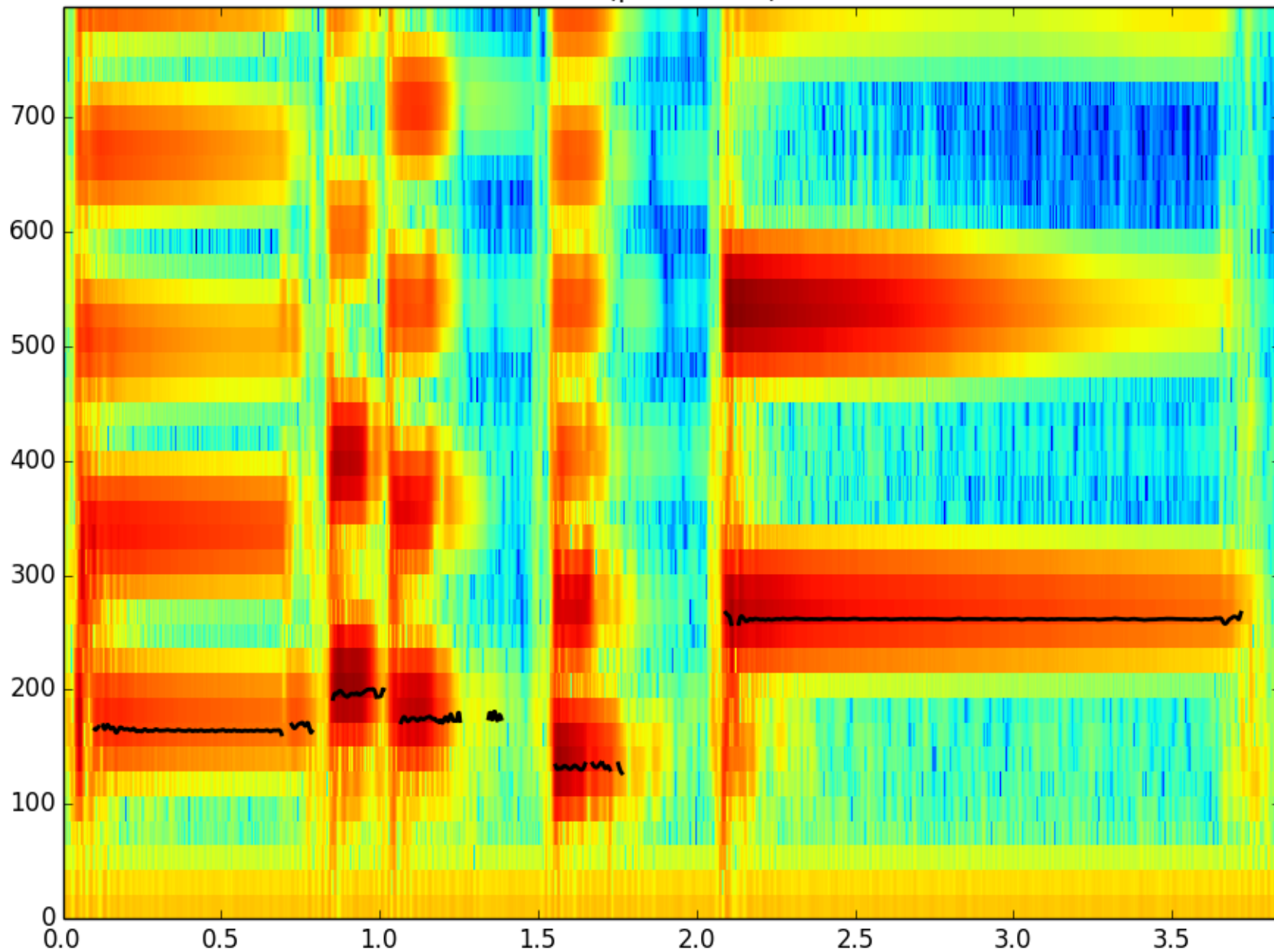
mX + peaks (oboe-A4.wav)



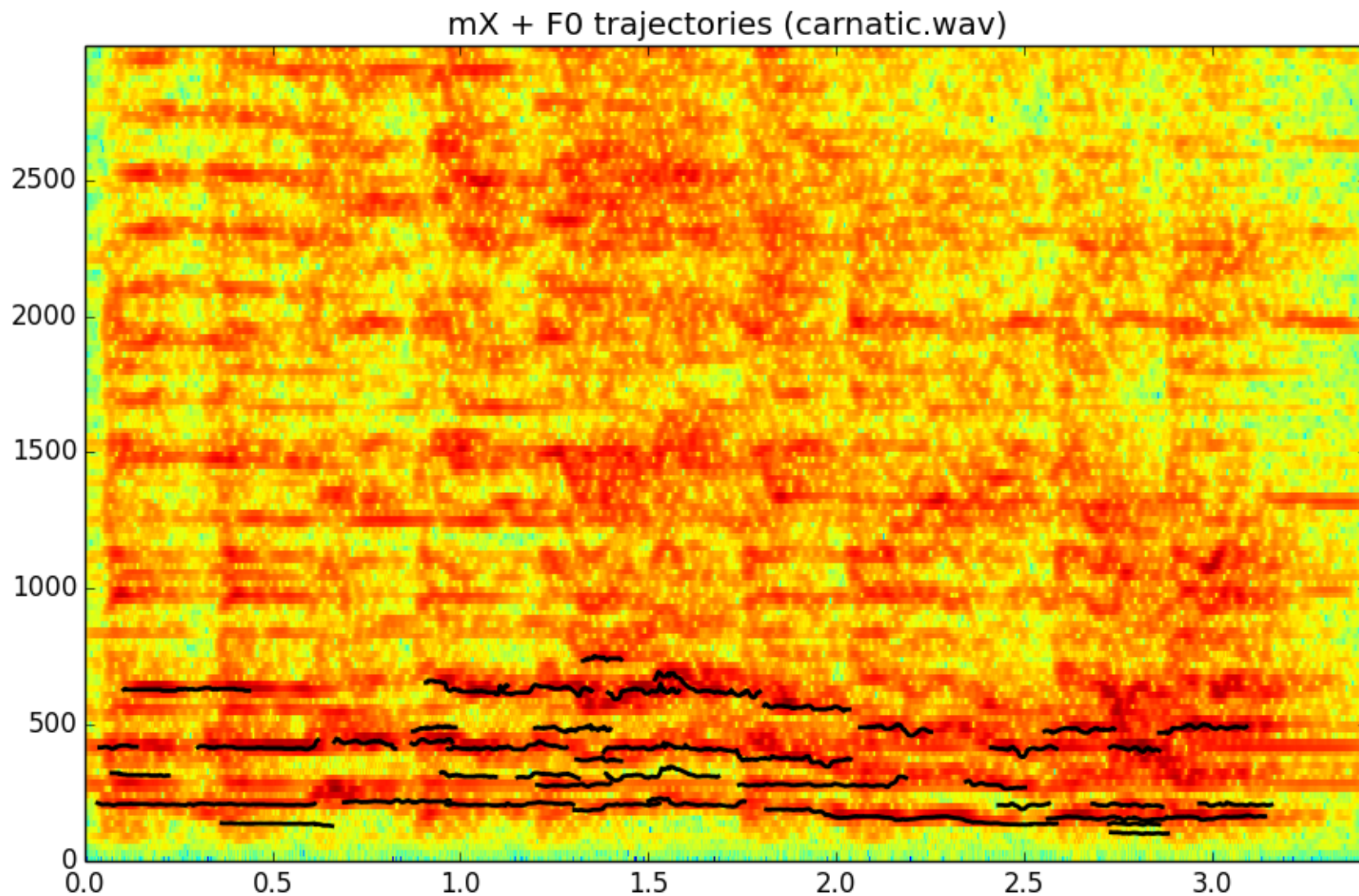
TWM Errors



mX + f0 (piano.wav), TWM

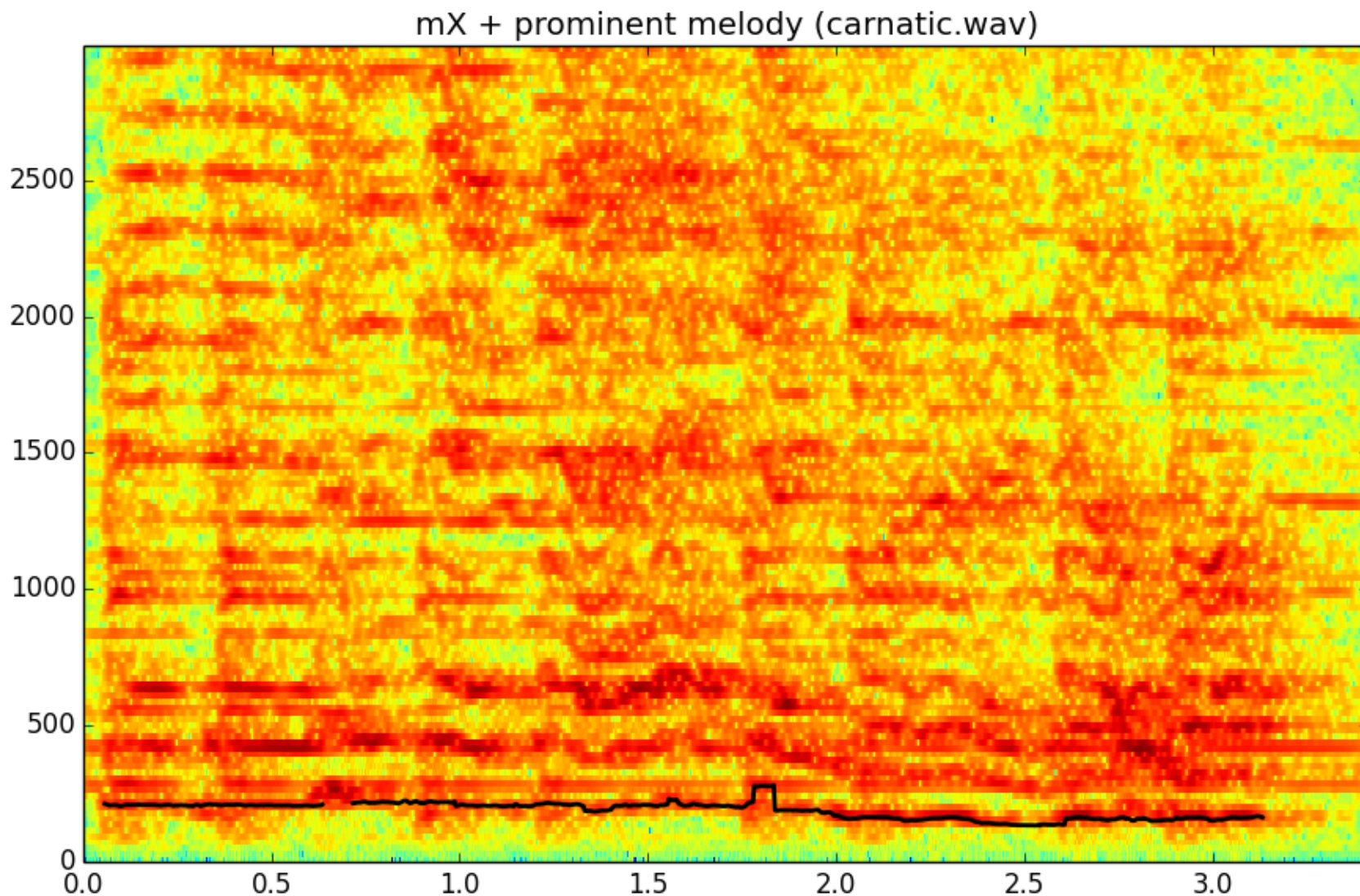


F0 in polyphonic signals



Prominent pitch in polyphonic signals

(Salamon and Gómez, 2012)



References and credits

- More information in:
http://en.wikipedia.org/wiki/Fundamental_frequency
http://en.wikipedia.org/wiki/Pitch_detection_algorithm
<http://en.wikipedia.org/wiki/Autocorrelation>
- F0 detection algorithms:
 - A. de Cheveigné and H. Kawahara. “YIN, a fundamental frequency estimator for speech and music,” J. Acoust. Soc. Am. 111, 1917 (2002).
 - R. C. Maher and J. W. Beauchamp, “Fundamental frequency estimation of musical signals using a Two-Way Mismatch procedure,” J. Acoust. Soc. Am., vol. 95., no. 4, pp. 2254-2263 (1994).
 - J. Salamon and E. Gómez, "Melody extraction from polyphonic music signals using pitch contour characteristics," IEEE Transactions on Audio, Speech, and Language Processing, vol. 20, no. 6, pp. 1759–1770 (2012).
- Sounds from: <http://www.freesound.org/people/xserra/packs/13038/>
- Slides released under CC Attribution-Noncommercial-Share Alike license and code under Affero GPL license; available from <https://github.com/MTG/sms-tools>

6T2: Fundamental frequency detection

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